PRD-01 · Onsite: the product

Status: Current (v0.1.2) · **Updated:** 2026-10-30 · **Owner:** Prateesh Sulikeri · **Feature PRDs:** 02–07

1. The problem

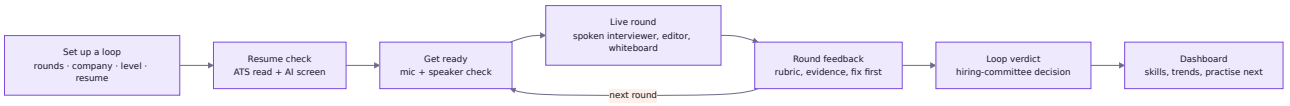
Software engineers prepare for interview loops by reading, grinding problems and, if they are lucky, doing a few mock interviews with friends. Real practice is hard to get:

- **Mock interviews with people are scarce.** They need a willing senior engineer, a time slot, and they rarely cover a whole loop.
- **Chat bots are not interviews.** Typing to a chat window does not train you to think out loud, take follow-ups, or manage the clock.
- **Generic practice misses the company.** Each company runs its own loop: different rounds, different bar, different style. That information is scattered across candidate reports.
- **Feedback is vague.** "Good job" does not tell you what a hiring committee would write.

Onsite started as a claude.ai artifact. It hit two walls: the sandbox blocks the microphone, and browser text-to-speech sounds robotic. Onsite is the native rebuild.

2. What Onsite is

A desktop app that runs realistic, spoken interview loops on your own computer, then tells you honestly how you did.



3. Who it is for

Persona	Situation	What they need from Onsite
The owner (primary)	Backend engineer: Java, Kafka, PostgreSQL, Redis, payments. Windows 11 + WSL. Has Claude Pro. Preparing for SDE-2 / senior loops.	Spoken rounds that push on real backend depth. Company loops built from real reports. Honest scores.
Friends (secondary)	Engineers at various levels. Some have no Claude subscription. Mostly Windows, some macOS/Linux.	Install in minutes, bring any AI (free tiers count), and get the same experience.

Persona	Situation	What they need from Onsite
Contributors (tertiary)	Developers who find the open-source repo.	Clear code, clear docs, a way to run it without the owner's setup.

4. Goals and non-goals

Goals

1. **Feel like a real interview.** A spoken conversation with a persona who follows up, pushes back, keeps time, and closes the round, at the level picked.
2. **Cover a whole loop.** Every common round type, run in order, with a verdict at the end.
3. **Company-true practice.** Research a company and role, then run *that* loop with its reported questions.
4. **Honest, actionable feedback.** Rubric scores with evidence from the transcript, the one thing to fix first, and what a strong answer would have said.
5. **Private and cheap.** Speech stays local. The user picks the AI and pays nothing extra to Onsite.

Non-goals

- **Not a hosted service.** No accounts, no server, no telemetry.
- **Not a question grinder.** The coding bank exists to make coding rounds real, not to replace LeetCode.
- **Not a cheating tool.** Nothing in Onsite helps during a real interview.
- **Not multi-user.** One person per install.
- **Not multilingual (yet).** Interviews are English only, on purpose: the speech models are tuned for it.

5. The rules Onsite never breaks

These come from `CLAUDE.md` → *Non-negotiables*. Every feature must respect them.

#	Rule	Why	What it means for a feature
N1	**Bring your own AI.** Claude Code CLI, or the user's own key for Claude API, OpenAI, Gemini or Groq. Keys stay in local SQLite. No SDKs.	Friends may not have Claude Pro. Free tiers (Gemini, Groq) must work.	Every model call must work on all five providers.
N2	**Runs natively on Windows**, owning the real mic and speakers.	The artifact sandbox blocked the mic.	No browser-only APIs for audio.

#	Rule	Why	What it means for a feature
N3	**Speech is local** (VoiceStudio), with fallbacks so the app never dead-ends.	Privacy, cost, latency.	Any voice failure must offer "Start typed".
N4	**The interviewer never gets tools.** Only the research job may search the web.	A model with tools in a live chat is a risk and a distraction.	New model calls send no tools unless they are research.
N5	**Model output and dossiers are data.** Never executed.	Prompt injection through a dossier or a reply.	Parse as JSON or show as text; nothing else.

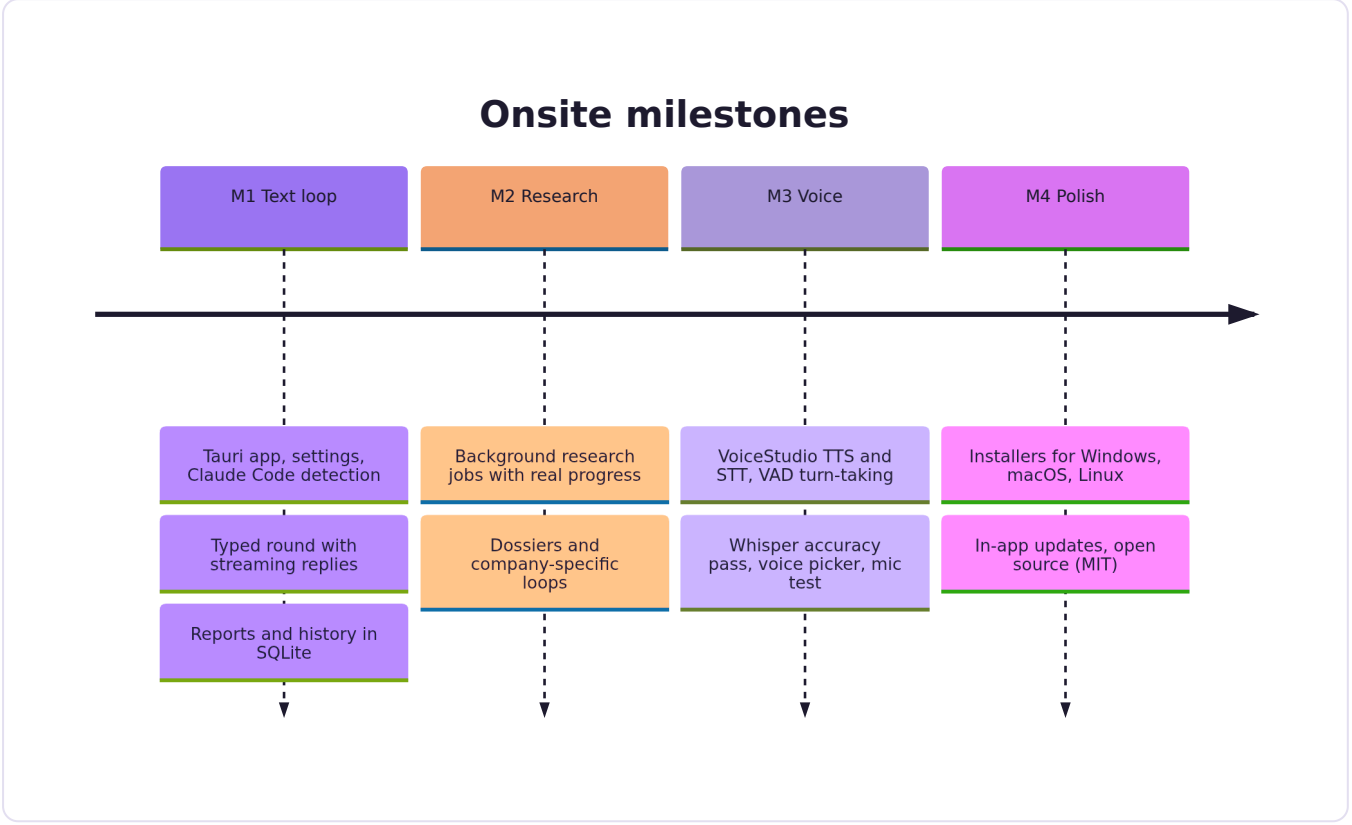
6. Features and status

Area	Feature	Status	PRD
Loops	General loops: 9 round types, 14 company styles, 5 levels, 3 temperaments	Shipped	PRD-02
Loops	Interviewer panels: 2 panels × 5 personas with voices and backstories	Shipped	PRD-02
Loops	Round feedback report and loop verdict	Shipped	PRD-02
Loops	Dashboard: skills, trends, weekly goal, practise next	Shipped	PRD-02
Loops	Code comprehension round redesign	Planned (research in `docs/comprehension-round-research.md`)	PRD-02
Voice	Spoken rounds via VoiceStudio, Whisper accuracy pass, mute, barge-in	Shipped	PRD-03
Voice	Typed rounds (fallback and by choice)	Shipped	PRD-03
Research	Company research jobs → dossiers; company-specific loops	Shipped	PRD-04
Research	Dossier import/export	Shipped	PRD-04
Resume	ATS parse check, six ATS profiles, keyword match	Shipped	PRD-05
Resume	AI screen (A–D grade against a posting)	Shipped	PRD-05
Coding	Editor with 6 languages; Run/Submit in Java, Python, C++, JS	Shipped	PRD-06

Area	Feature	Status	PRD
Coding	Whiteboard (Excalidraw) described to the interviewer as text	Shipped	PRD-06
App	Five AI providers, live model lists, health check	Shipped (Claude API, OpenAI and Gemini not yet run against a live key)	PRD-07
App	Themes: Onsite, Nord, Rosé Pine, Gruvbox; light/dark; custom backdrop	Shipped	UI design
App	In-app updates from GitHub releases	Shipped	PRD-07
App	Storage limits and cleanup	Shipped	PRD-07
App	Code-signed Windows installer (SignPath)	Pending: application submitted	DD-08
App	MCP server for Claude Code in a terminal	Planned (phase 3)	—

7. Milestones

Each milestone had to be usable on its own.



Milestone	Acceptance test	Result
M1	A 10-minute typed DSA round completes end to end and the report renders.	Met
M2	A new company dossier is produced without touching a chat window.	Met
M3	A full spoken round with no typing; first audio < 2.5 s after the user stops talking.	Spoken round met. Latency: last-segment STT ~3.4 s on CPU Parakeet; Whisper on GPU 1.1–1.5 s per answer. The 2.5 s budget is tight and still being tuned.
M4	Installers built by CI on tag push; README screenshots.	Installers met. macOS/Linux builds not yet run by a person.

8. How we know it works

There is no telemetry, so success is measured by the owner and by friends' reports.

Signal	Target	How it is measured
Time to first interviewer audio	< 2.5 s after the candidate stops	`voice {kind:"latency"}` events, logged per turn
Time to first token (typed)	< 2 s on API providers	`ttft_ms` on every interviewer turn, saved in the transcript
Rounds that end cleanly	Interviewer closes with `[[END]]`, report renders	History; "not enough to review" rate
Transcript accuracy	Tech terms right (Redis, idempotency, Postgres)	Spot checks; Whisper vs Parakeet comparisons
Friends can install	From download to first round in < 15 min	Friends' reports

9. Constraints

- **Claude Code has no temperature flag.** Variety comes from rotating "openers" and showing the model how recent rounds began (see DD-02).
- **Free tiers are rate-limited.** Groq gives about 8K tokens/minute. A coding turn sends about 5.4K. Every provider waits out a 429 (up to 60 s per wait, 90 s total).
- **VoiceStudio is a separate app.** Onsite can launch it and check its version, but never installs or updates it.
- **The owner's GPU** (RTX 5050) keeps Whisper large-v3 loaded. Machines without a GPU fall back to Parakeet text.

10. Open questions

- How should the comprehension round work? (Research done: real repos vs curated small repos vs bug datasets.)
- Should rounds support languages other than English when better speech models arrive?
- Should there be an "AI assistant" switch so the candidate can practise rounds that allow AI help?
- Which parts should the planned MCP server expose, beyond read-only dossiers and reports?